

Assignment 03

Length of life (LoL) and quality of life (QoL) are both derived from the value the authors attribute to the DALY (\$87,500) with a ratio of YLD to YLL of .30 (pg. 24). The value of deaths avoided due to falling tobacco consumption and the value of QoL gained are likely highly related as a reduction in deaths and disabilities coincide with a reduction in tobacco consumption overall. Recalculating the NPV and BCR for each value independently will allow for an initial exploration of these potential inflations.

The recalculations (Table 1 in the Excel file) clearly show that the years of life gained by an avoided death make up a large portion of the stated benefits and thus omitting them greatly affects the BCR of the proposal – now indicating a negative net present value and BCR of .69. This makes intuitive sense given the stated ratio of .30 for those ‘individual’ benefits.

To further evaluate for potential inflation of the data, the values in Table 7.1 can be recalculated based on the authors’ stated assumptions and cited research. First, deaths avoided are interpolated from Table 5.2. Second, to assign value these avoided deaths, we multiply by the DALY value (87,500) and the years of life gained (9) to arrive at an aggregate value in each period. Last, we compare those values against the combined LoL and QoL gains from Table 7.1 in the RIS (Table 2 in Excel Calculations File). The benefits in Table 7.1 of the RIS are, on average, \$10m higher than what the stated assumptions would imply.

To confirm this overstatement, the assumption values (87,500 per DALY and 9 years of life gained) can be extracted from the data in Table 7.1 of the report. Table 3 in the Excel file divides the stated benefits in Table 7.1 of the report by the deaths avoided from Table 5.2 of the report. The result is a value that, when divided by 9 (years of life gained) should yield the 87,500 per DALY and when divided by the value of the DALY should yield 9 years. However, Table 4 in the Excel file shows that both stated values are inflated in the table. Finally, Table 5 shows that when we divided the benefits from Table 7.1 of the report by the authors’ assumptions, we arrive at ‘deaths avoided’ values that are also higher than the interpolated values from Table 5.1.

I cannot locate anything in the methodology that would indicate why the values in Table 7.1 exceed what is implied in the stated assumptions and cited data. I have learned that gleaning clear methodology is sometimes difficult and that even the most official reports should be scrutinized instead of accepting results at face value. It is also clear that any attempt at evaluating the soundness of a given study requires a willingness to engage with the underlying material, even when it may be complex or unfamiliar.